SUPPLY CHAIN MANAGEMENT FOR ENGINEERS

MINI PROJECT REPORT

Course Code: UE23CS342BA1
Problem Statement: P6 – Supplier Relationship Management

Professor: Prof. Raghu B. A Rao
Submission Date: 20/04/2026

TEAM MEMBERS

- PES1UG23CS662 – Ramjhith VC
- PES1UG23CS233 – Harsh Prabhakar K

---

1. PROBLEM STATEMENT ANALYSIS

1.1 Problem Definition

Supplier Relationship Management (SRM) is a critical component of supply chain operations. Organizations face challenges such as lack of visibility into supplier performance, inconsistent delivery timelines, poor communication, and quality issues.

The assigned problem focuses on:
Inefficient supplier relationship management due to lack of collaboration tools, tracking systems, and performance monitoring, leading to delays, increased risks, and higher costs.

1.2 Scope of the Project

This project addresses the following aspects:

- Supplier performance monitoring
- Procurement efficiency
- Quality tracking
- Risk prediction and analysis

1.3 SCM Areas Covered

- Procurement Management
- Supplier Performance Analytics
- Quality Control
- Risk Management
- Logistics and Delivery Monitoring

1.4 Proposed Solution

An **AI-powered Supplier Relationship Management Dashboard** that:

- Tracks supplier KPIs in real time
- Provides interactive visual insights
- Uses Machine Learning for prediction
- Enables proactive decision-making

---

2. DASHBOARD DESIGN & DEVELOPMENT

2.1 Overview

The dashboard is developed using:

- Python (Dash / Flask)
- Plotly for visualization
- SQLite database

It provides a clean and interactive interface for supply chain managers.

2.2 Key Features

- Supplier filtering
- Date range filtering
- Order status filtering
- KPI selection
- Interactive charts
- Real-time updates

2.3 Key Performance Indicators (KPIs)

- On-Time Delivery Rate (%)
- Defect Rate (%)
- Average Lead Time (days)
- Lead Time Variability
- Cost Variance (%)
- Supplier Risk Score (0–100)
- PO Acknowledgment Rate (%)

- Response Time to Queries

2.4 Visualizations Used

- Bar Charts → Supplier comparison
- Line Charts → Trend analysis
- KPI Cards → Quick summary
- Tables → Operational tracking

---

3. DATA STRUCTURE AND SAMPLE DATA

3.1 Database Design

A relational database using SQLite is implemented to store supply chain data efficiently.

3.2 Tables Description

Suppliers

- supplier_id (Primary Key)
- name
- location
- contact

PurchaseOrders

- po_id (Primary Key)
- supplier_id (Foreign Key)
- order_date
- expected_delivery_date
- actual_delivery_date
- cost
- quantity

Shipments

- shipment_id (Primary Key)
- po_id (Foreign Key)
- shipment_date
- delivery_status

- delay_days

QualityRecords

- quality_id (Primary Key)
- supplier_id (Foreign Key)
- defect_rate
- rejection_reason

CommunicationLogs

- log_id (Primary Key)
- supplier_id (Foreign Key)
- response_time
- communication_type

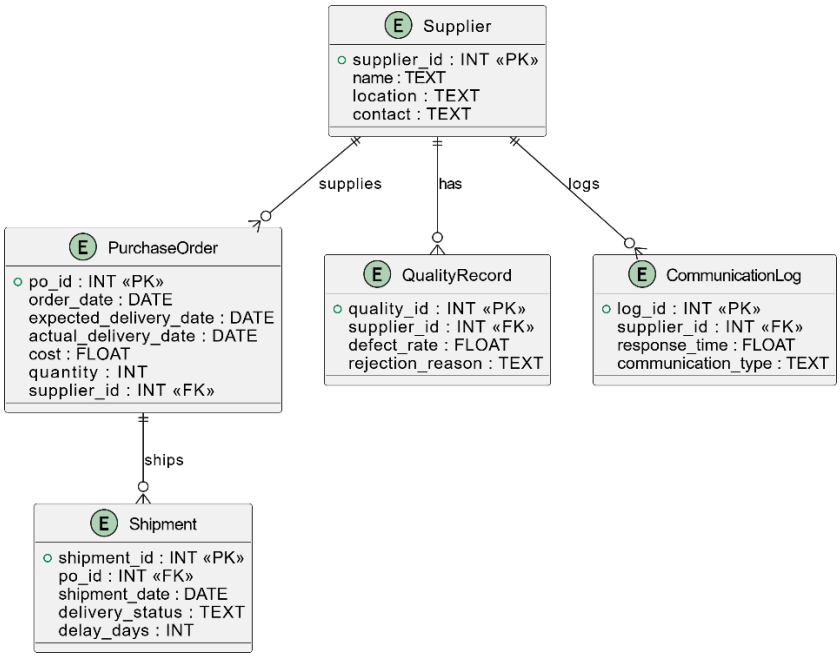
3.3 Sample Data

The system generates realistic sample data including:

- 12 suppliers
- 100 purchase orders
- Approximately 35% delayed deliveries
- Cost and defect variations

---

4. ER DIAGRAM (DATA RELATIONSHIP SCHEMA)



4.1 Relationships

- Supplier → PurchaseOrders (1:N)
- PurchaseOrders → Shipments (1:N)
- Supplier → QualityRecords (1:N)
- Supplier → CommunicationLogs (1:N)

4.2 Justification

The schema:

- Ensures proper data organization
- Supports KPI calculations
- Enables ML model training
- Reflects real-world supply chain systems

---

5. MACHINE LEARNING INTEGRATION

Machine Learning models are used to enhance decision-making and predict risks.

Assumptions made:

- Data is simulated
- Suppliers are independent
- No external disruptions modelled.

5.1 Delay Prediction Model

- Algorithm: Random Forest
- Input Features: supplier performance, lead time, historical delays
- Output: Probability of delivery delay

Performance:

- Accuracy: 0.588
- Precision: 0.818
- Recall: 0.643

Model Justification:
Random Forest is chosen because it handles nonlinear relationships effectively and works well with structured data. It is robust to noise and can capture complex interactions between supplier performance and delivery delays.

Result Interpretation:
The model achieves moderate accuracy (0.588), indicating that predictions are somewhat reliable but limited by the small and synthetic dataset. The high precision suggests that when a delay is predicted, it is often correct, making it useful for risk flagging.

5.2 Defect Risk Prediction

- Algorithm: Logistic Regression
- Input Features: supplier quality metrics, defect history
- Output: Probability of high defect rate

Performance:

- Accuracy: 0.886
- Precision: 0.333
- Recall: 0.111

Model Justification:
Logistic Regression is chosen due to its simplicity, interpretability, and effectiveness in binary classification problems such as defect risk prediction.

Result Interpretation:
Although the model shows high accuracy, the low recall indicates that many high-risk defect cases are not identified. This suggests the model is conservative and may require further tuning or more data.

5.3 Lead Time Prediction

- Algorithm: Random Forest
- Output Classes: Early / On-Time / Late

Performance:

- Accuracy: 0.353
- Precision: 0.471
- Recall: 0.353

Model Justification:
Random Forest is suitable for multi-class classification problems and is effective in identifying complex patterns in structured data.

Result Interpretation:
The low accuracy indicates that the model struggles to distinguish between early, on-time, and late deliveries, primarily due to limited and synthetic data. However, it still provides directional insights for planning.

5.4 Benefits

- Early risk detection
- Improved supplier selection
- Reduced delays and defects
- Better procurement planning

---

6. METHODOLOGY

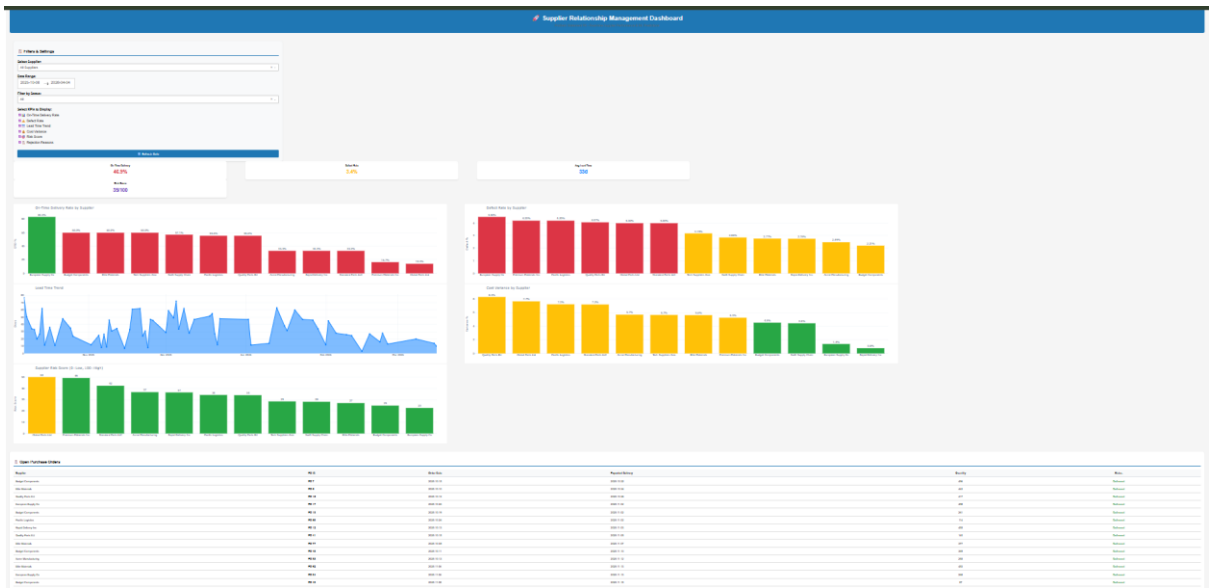
Steps followed in the project:

1. Understanding the problem
2. Designing the database
3. Generating sample data
4. Implementing KPI calculations
5. Developing the dashboard
6. Integrating ML models
7. Testing and validation

---

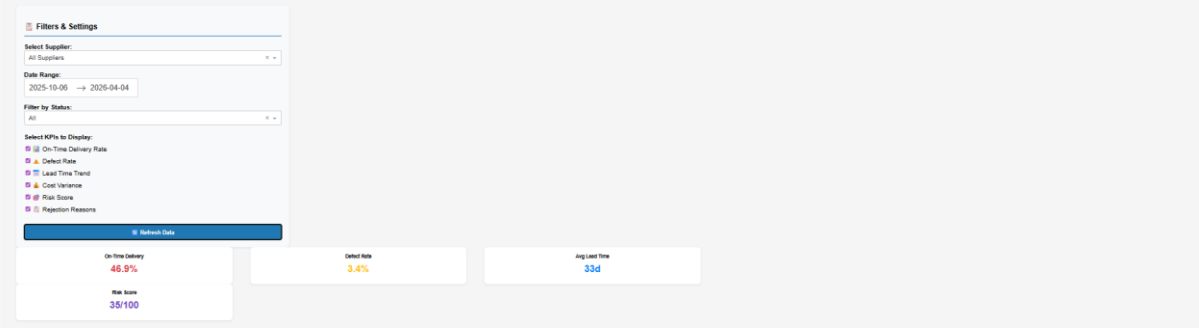
7. DASHBOARD SCREENSHOTS

7.1 Main Dashboard View



Description:
Displays an overview of supplier performance including KPI cards, charts, and filters. Provides a centralized view for decision-making.

7.2 KPI Cards Section

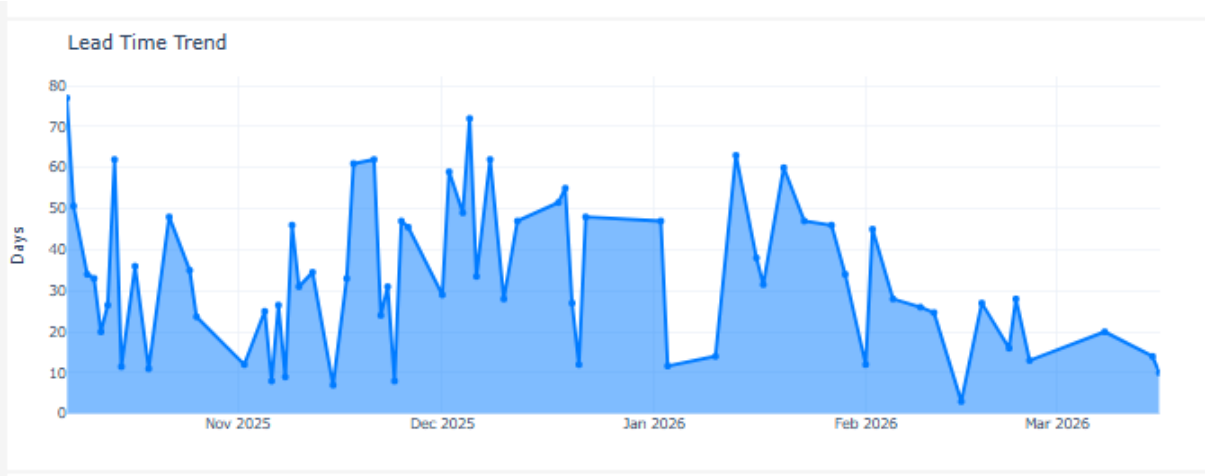


Description:
Shows key metrics such as On-Time Delivery Rate, Defect Rate, and Risk Score for quick performance assessment.

7.3 Supplier Comparison (Bar Chart)

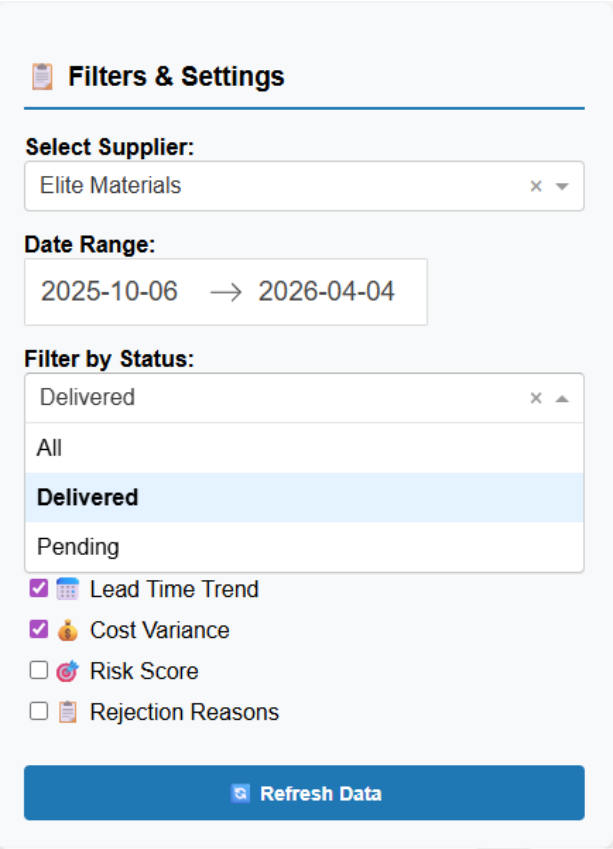
Description:
Compares suppliers based on KPIs such as defect rate and cost variance, helping identify top and underperforming suppliers.

7.4 Trend Analysis (Line Chart)



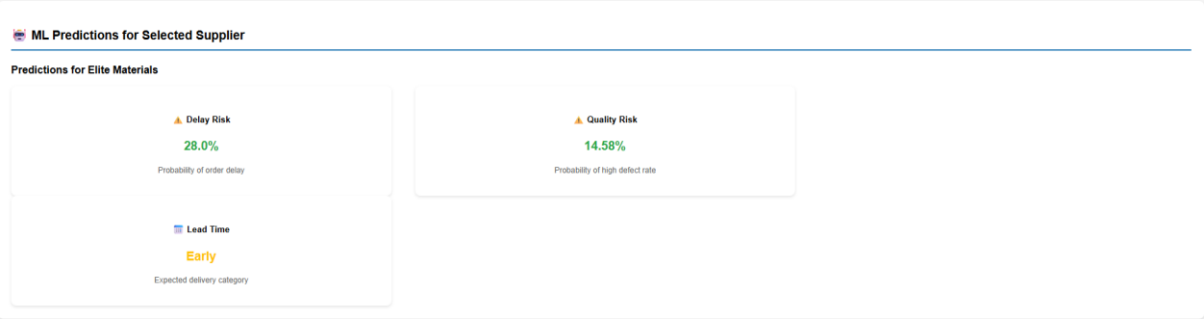
Description:
Shows trends in delivery performance over time, enabling identification of patterns and seasonal variations.

7.5 Filters Panel



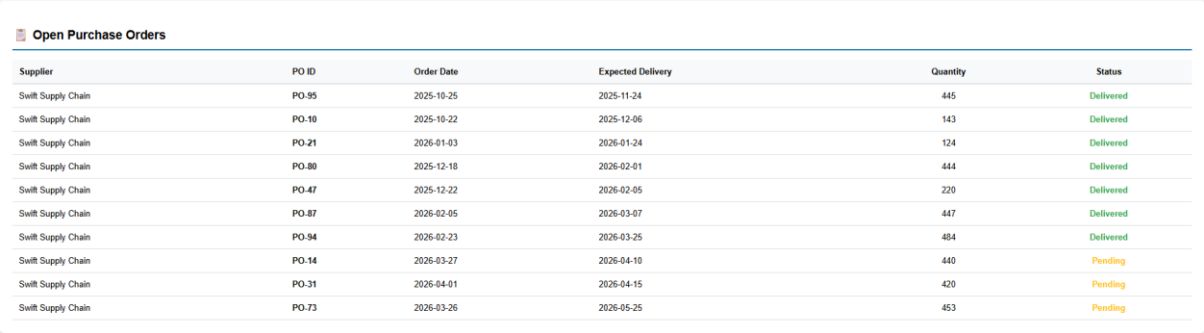
Description:
Allows filtering by supplier, date range, and order status to perform targeted analysis.

7.6 ML Prediction Section



Description:
Displays predicted delay probability and defect risk, enabling proactive decision-making.

7.7 Open Purchase Orders Table

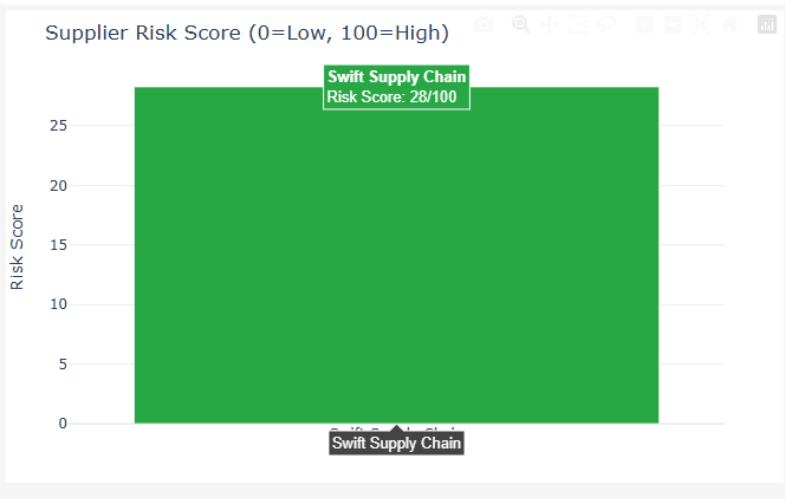


Open Purchase Orders

Supplier	PO ID	Order Date	Expected Delivery	Quantity	Status
Swift Supply Chain	PO-95	2025-10-25	2025-11-24	445	Delivered
Swift Supply Chain	PO-10	2025-10-22	2025-12-06	143	Delivered
Swift Supply Chain	PO-21	2026-01-03	2026-01-24	124	Delivered
Swift Supply Chain	PO-80	2025-12-18	2026-02-01	444	Delivered
Swift Supply Chain	PO-47	2025-12-22	2026-02-05	220	Delivered
Swift Supply Chain	PO-87	2026-02-05	2026-03-07	447	Delivered
Swift Supply Chain	PO-94	2026-02-23	2026-03-25	484	Delivered
Swift Supply Chain	PO-14	2026-03-27	2026-04-10	440	Pending
Swift Supply Chain	PO-31	2026-04-01	2026-04-15	420	Pending
Swift Supply Chain	PO-73	2026-03-26	2026-05-25	453	Pending

Description:
Shows detailed operational data of active orders, including delays and supplier information.

7.8 Risk Score Visualization



Description:
Represents supplier risk levels visually, helping managers quickly identify high-risk suppliers.

---

8. INSIGHTS DERIVED

- Suppliers with On-Time Delivery below 75% show higher delay probability
- High defect rates indicate poor supplier quality
- Cost variance reveals pricing inefficiencies
- Risk score effectively identifies unreliable suppliers

---

9. BUSINESS IMPACT

Aspect	Before	After
Visibility	Low	High
Decision Making	Reactive	Proactive
Risk Handling	Delayed	Predictive
Supplier Selection	Manual	Data-driven

---

10. CHALLENGES FACED

- Generating realistic sample data
- Machine learning model tuning
- Integration of multiple components
- Dashboard responsiveness

---

11. LIMITATIONS

- Uses simulated data instead of real-world data
- Limited dataset size affects ML accuracy
- No real-time system integration

---

12. FUTURE IMPROVEMENTS

- Real-time data integration
- Advanced forecasting models

- Supplier recommendation system
- Automated alerts and notifications

---

13. CONCLUSION

This project successfully addresses Supplier Relationship Management challenges using data visualization, KPI monitoring, and Machine Learning techniques.

The developed system enables **efficient, predictive, and data-driven supply chain management**, improving supplier performance and overall operational efficiency.

Project GitHub Link: https://github.com/ramjhith/Supplier-Dashboard.git

---